CHAPTER 7

ORGANISMS AND POPULATIONS

CHAPTER ESSENCE AT A GLANCE:
Population growth kinetics, age pyramids, and the multi-species matrix of interspecific ecological interactions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Population Attributes and Dynamic Density:** Populations possess group attributes: population density (N), natality rate (B, births per capita), mortality rate (D, deaths per capita), immigration (I), and emigration (E). Population growth is modeled by N(t+1) = N(t) + [(B + I) - (D + E)]. Age pyramids display age distribution: expanding (broad base of pre-reproductive individuals), stable (bell-shaped with uniform pre-reproductive and reproductive cohorts), or declining (urn-shaped with narrow base).

• **Exponential versus Logistic Population Growth:** When resources are unlimited, populations grow exponentially yielding a J-shaped curve modeled by dN by dt = rN (integral form Nt = N0 e^{r}t, where r is the intrinsic rate of natural increase). Because natural resources are finite, real populations follow the Verhulst-Pearl logistic growth equation: dN by dt = rN [(K - N) over K], producing an S-shaped sigmoid curve with carrying capacity K.

• **Harmful Interspecific Interactions: Competition and Predation:** Competition (-/-): Both interacting species are harmed. Gause's Competitive Exclusion Principle states that two closely related species competing for identical limiting resources cannot coexist indefinitely; MacArthur demonstrated that competing warblers coexist through resource partitioning. Predation (+/-): Predators act as conduits of energy transfer and maintain community diversity by preventing prey species from dominating.

• **Beneficial and Commensal Interactions:** Mutualism (+/+): Both species benefit (e.g., lichens pairing algae and fungi, mycorrhizae, fig tree pollinated obligately by specific fig wasp). Commensalism (+/0): One species benefits while the other is unaffected (e.g., epiphytic orchid on a mango branch, barnacles on whale skin, cattle egret foraging near grazing cattle, sea anemone and clown fish). Amensalism (-/0): One species is harmed while the other is unaffected (e.g., Penicillium inhibiting bacterial growth).

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right) \quad [K = \text{Carrying Capacity},\ r = \text{Intrinsic Rate of Increase}]$

Verhulst-Pearl Logistic Growth Sigmoid Formulation

$\frac{dN}{dt} = rN \rightarrow N_t = N_0 e^{rt} \quad [\text{J-shaped curve under unlimited resources}]$

Exponential Geometric Population Growth Model

$N_{t+1} = N_t + [(B + I) - (D + E)] \quad [B = \text{Births},\ D = \text{Deaths},\ I = \text{Immigration},\ E = \text{Emigration}]$

Demographic Population Density Dynamic Equation

$\text{Interactions} = \text{Mutualism}\ (+,+) + \text{Competition}\ (-,-) + \text{Predation}\ (+,-) + \text{Commensalism}\ (+,0)$

Ecological Interspecific Interaction Matrix

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Abiotic Factors, Responses to Abiotic Factors (Regulators and Conformers), and Adaptations were completely deleted under the rationalized syllabus.

• **Alert:** In the logistic equation, when N equals K, (K - N) over K becomes zero, and the population growth rate dN by dt drops to zero, establishing a stable plateau.

• **Alert:** Brood parasitism in birds (e.g., cuckoo laying eggs in crow's nest) is a parasitic interaction where the host incubates the eggs, showing co-evolution of egg mimicry.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Write the Verhulst-Pearl logistic growth equation. Explain what each variable in the equation represents. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Equation: dN by dt = rN * $\frac{K-N}{K}$. **[1 Mark]**

• Variables: N = Population density at time t; r = Intrinsic rate of natural increase; K = Carrying capacity of the environment. **[1 Mark]**

2. What is Gause's Competitive Exclusion Principle? How did MacArthur demonstrate that competing species can avoid competitive exclusion? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Gause's Principle: Two closely related species competing for the same limiting resources cannot coexist indefinitely, and the competitively inferior species will eventually be eliminated. **[1 Mark]**

• MacArthur's finding: Showed that five closely related species of warblers living on the same tree avoided competition and coexisted through resource partitioning by foraging at different heights and times. **[1 Mark]**

3. In a pond, there are 20 lotus plants initially. Over one year, 8 new lotus plants are born through vegetative reproduction while 4 die. Calculate the birth rate, death rate, and net population growth of the lotus population. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Birth rate: Number of births divided by initial population = 8 divided by 20 = 0.4 offspring per lotus plant per year. **[1 Mark]**

• Death rate: Number of deaths divided by initial population = 4 divided by 20 = 0.2 deaths per lotus plant per year. **[1 Mark]**

• Net population growth: Net increase = Births - Deaths = 8 - 4 = +4 plants; final population = 20 + 4 = 24 lotus plants. **[1 Mark]**

4. Define the following ecological interactions and provide an authentic natural example for each: (a) Commensalism, (b) Mutualism, and (c) Amensalism. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Commensalism (+/0): One species benefits while the other is neither harmed nor benefited (e.g., an orchid growing as an epiphyte on a mango tree branch). **[1 Mark]**

• (b) Mutualism (+/+): Both interacting species derive positive mutual survival benefits (e.g., symbiotic mycorrhizal association between fungi and roots of higher plants). **[1 Mark]**

• (c) Amensalism (-/0): One species is harmed or inhibited while the other species is unaffected (e.g., Penicillium mold secreting penicillin antibiotic that inhibits bacteria without gaining direct benefit). **[1 Mark]**

5. Describe the three types of age pyramids representing expanding, stable, and declining human populations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Expanding population pyramid: Triangular shape with a very broad base; contains a high proportion of pre-reproductive individuals compared to reproductive and post-reproductive cohorts, indicating rapid population growth. **[1 Mark]**

• Stable population pyramid: Bell-shaped; pre-reproductive and reproductive age groups are almost equal in size, indicating that population size remains nearly stationary. **[1 Mark]**

• Declining population pyramid: Urn-shaped with a narrow base; the pre-reproductive group is smaller than the reproductive group, indicating negative population growth. **[1 Mark]**

6. Explain how predators act as conduits for energy transfer and maintain species diversity in ecological communities. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Conduits of energy transfer: Predators feed on herbivores and lower carnivores, transferring photosynthetic biomass and energy across successive trophic levels to top carnivores. **[1.5 Marks]**

• Maintaining species diversity: Predators control prey population densities, preventing competitively dominant prey species from outcompeting and eliminating weaker prey species (e.g., removal of predatory starfish Pisaster caused extinction of 10 invertebrate prey species). **[1.5 Marks]**

7. Contrast exponential (J-shaped) population growth with logistic (sigmoid S-shaped) growth. Explain why logistic growth is considered biologically more realistic in natural ecosystems. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Exponential growth: Occurs when resources (food, space, shelter) are unlimited in the habitat; population grows exponentially without environmental resistance, yielding a steep J-shaped curve modeled by dN by dt = rN. **[1.5 Marks]**

• Logistic growth: Occurs when environmental resources are finite; population exhibits an initial lag phase, followed by acceleration (log phase), and deceleration as density approaches carrying capacity K, yielding an S-shaped sigmoid curve modeled by dN by dt = rN * $\frac{K-N}{K}$. **[1.5 Marks]**

• Biological realism: In real nature, no habitat possesses limitless food and space; competition for scarce resources sets an upper sustainable carrying capacity K beyond which populations cannot grow without mortality spikes. **[1 Mark]**